Problem 1

$$f(x,y)=\frac{xy}{x+y}, \qquad x,y>0 \quad \text{(parts 1–2).}$$

1. Find the linearization $L(x,y)$ of f at $(6,3)$.
2. Use the linearization L from part (1) to approximate $f(6.12, 2.94)$.
3. For $z=g(x,y)=x^2+xy+2y^2$, find the point on the surface where the tangent plane is parallel to $3x+5y-z=0$, and find the equation of that tangent plane.

Problem 2

$$z = f(x, y) = x^2 + 2xy + 3y^2, \qquad P = (1, 1, 6).$$

1. Find the differential dz.
2. When (x, y) changes from $(1, 1)$ to $(1.02, 0.99)$, compute dz and the exact change Δz. Does $dz = 0$ mean that z does not change?

Problem 3

$$w = f(x, y, z) = x^2 e^z + xy, \qquad P = (1, 1, 0).$$

1. Find f_x, f_y, f_z, and evaluate them at P.
2. Let $x = e^{2u}$, $y = 1 - u$, and $z = 3u$. Use the Chain Rule to find dw/du at $u = 0$.
3. Set $x = s + t$, $y = s - t$, $z = st$, and $W(s, t) = f(x, y, z)$. Use the Chain Rule to find $W_s(1, 0)$ and $W_t(1, 0)$.

Problem 4

$$S: \quad x^2 + xy + 2y^2 + z^2 = 8, \qquad P = (1, 1, 2).$$

1. Near P, regard z as a function of x, y. Use implicit differentiation to find z_x, z_y, and evaluate them at P.
2. A point moves on S with $x = 1 + u$, $y = e^u$, and $z(0) = 2$. Find dz/du at $u = 0$.
3. Set $x = s + t$, $y = s^2 - t$. At $(s, t, z) = (1, 0, 2)$, find $\partial z/\partial s$ and $\partial z/\partial t$.